Nokia Core Node BLR-CORE-02 Routing Protocol Flap SOP - Alarm Code: ALARM_9533

Node: BLR-CORE-02 | **Vendor:** Nokia | **Severity:** Major | **Alarm Code:** ALARM_9533

Description

This Standard Operating Procedure (SOP) is designed to address a critical issue on the Nokia Core Node BLR-CORE-02, which has resulted in a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) alarm code of ALARM_9533. This failure domain occurs due to the complex interaction between the routing protocols and the node's core architecture. In a Nokia Core Node, the Border Gateway Protocol (BGP) and Open Shortest Path First (OSPF) protocols are used to exchange routing information with neighboring nodes. However, when there is a disruption in the routing protocol operation, such as a BGP route churn or OSPF adjacency loss, it can cause the ISIS LSP flooding, leading to a major alarm. This can be caused by various factors, including misconfiguration, hardware failures, or software bugs. The root cause of this issue needs to be identified and addressed to prevent further disruptions to the network. Understanding the underlying protocol mechanics is crucial in resolving this issue. The BGP route churn occurs when there is a change in the routing table, causing the node to advertise new routes or withdraw existing ones. This can lead to OSPF adjacency loss, which in turn causes the ISIS LSP flooding. The OSPF adjacency loss occurs when there is a disruption in the OSPF protocol operation, such as a neighbor relationship loss. This can be caused by various factors, including misconfiguration, hardware failures, or software bugs. The ISIS LSP flooding occurs when there is an excessive number of LSPs (Link-State Packets) being flooded on the network, causing congestion. This can be caused by various factors, including misconfiguration, hardware failures, or software bugs. In this SOP, we will identify the root cause of the issue, perform diagnostic checks, and implement resolution steps to address the problem.

Pre-requisite Checks

- Ensure that the Nokia Core Node BLR-CORE-02 is accessible and can be reached from the network operations center.
- Verify that the redundant link status is stable and not causing any disruptions to the network.
- Check that the site location is unknown and that the remote hands are dispatched to the site.

Potential Root Causes

Cause	Probability
Misconfigured BGP neighbor relationship	High
Hardware failure in the OSPF protocol implementation	Medium
Software bug in the ISIS protocol implementation	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp neighbor, Expected Output: Neighbor AS Up/Down
State/ MsgRcvd MsgSent OutQ Last Error Suppress, Description: This
command will display the BGP neighbor relationships, including the
state and the number of messages received and sent. This will help
identify if there are any issues with the BGP neighbor relationship.

Command: show ospf neighbor, Expected Output: OSPF Neighbor ID Pri
State Dead Time Address Interface, Description: This command will
display the OSPF neighbor relationships, including the state and the
dead time. This will help identify if there are any issues with the
OSPF neighbor relationship.

Command: show isis lsp, Expected Output: LSP LSP ID Seq Age LSP Metric
Next Hop LSP Type LSP Length, Description: This command will display
the ISIS LSPs, including the sequence number, age, and metric. This
will help identify if there are any issues with the ISIS LSP flooding.
```

Resolution Steps

Step 1: Step: 1. Check the BGP neighbor relationship, Command: show bgp neighbor, Description: Check the BGP neighbor relationship to ensure that it is stable and not causing any disruptions to the network.

Step 2: Step: 2. Check the OSPF neighbor relationship, Command: show ospf neighbor, Description: Check the OSPF neighbor relationship to ensure that it is stable and not causing any disruptions to the network.

Step 3: Step: 3. Check the ISIS LSP flooding, Command: show isis lsp, Description: Check the ISIS LSP flooding to ensure that it is not excessive and not causing any disruptions to the network.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP Neighbor ID AS Route
Refcnt Stabilized, Description: This command will display the BGP
summary, including the number of routes and the stabilized count. This
will help verify if the BGP neighbor relationship is stable.

Command: show ospf database, Expected Output: OSPF Database ID Type
Metric Next Hop, Description: This command will display the OSPF
database, including the type and metric. This will help verify if the
OSPF neighbor relationship is stable.

Command: show isis database, Expected Output: ISIS Database ID Type
Metric Next Hop, Description: This command will display the ISIS
database, including the type and metric. This will help verify if the
ISIS LSP flooding is not excessive.
```

Roll-back Procedures

- Step: 1. Restore the original configuration, Command: copy running-config startup-config, Description: Restore the original configuration by copying the running configuration to the startup configuration.

- Step: 2. Reboot the node, Command: reboot, Description: Reboot the node to ensure that it is restored to its original state.